National Sun Yat-Sen University

SOPC 設計實務及 FPGA 系統整合設計

Homework #3
Due April 17 2026 9:00AM

1. **<Code study 1>** Trace the entire code in the enclosed design file ***HW3_1.rar*** which was originally used in an SoC development board based on the AMBA 2.0 protocol. You are required to report your findings by drawing a block diagram and explaining the function of the design. Additionally, describe the detailed operation of the enclosed RGV-to-YUV conversion IP. The score for this assignment will be based on the level of detail you provide in the block diagram and how well we believe you understand the design. For example, you could illustrate key internal AMBA bus signal interconnections and demonstrate how multiple masters and slaves can be realized in an FPGA with limited signal ports.

2. **<Code study 2>** Study the two Verilog files enclosed in ***HW3_2.rar***. You should trace the design described by these two files and report on the following issues:

   (1) Provide a general description of the circuit and its function as described by these files.

   (2) Draw the ***datapath*** of the circuit implemented by these designs. (You do not need to include signals primarily used for control. Focus on showing the data flow involving the following signals: ***s00_axi_awaddr***, ***s00_axi_araddr***, ***s00_axi_wdata***, and ***s00_axi_rdata***.)

   (3) Describe the function of the parameters ***ADDR_LSB***, and ***OPT_MEM_ADDR_BITS***.

   (4) Provide the memory mapped information of this design if ***C_S_AXI_DATA_WIDTH*** is set to 64.

   (5) Explain the purpose of the following code segments used in the provided code.

```
for ( byte_index = 0; byte_index <= (C_S_AXI_DATA_WIDTH/8)-1;
byte_index = byte_index+1 )
```